ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE G

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Multivibrators, SCRs and inductors, electric drives, servo feedback and op-amp controllers, hydraulic valves, accumulators and a log-splitter circuit, electro-hydraulic interlocks, and fault-finding on pumps and power packs.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all circuits and symbols neatly, using a ruler and standard symbols.
5. Write neatly and legibly.

QUESTION 1

1.1 Name the THREE types of multivibrators and give ONE use of each. (6)

1.2 Explain how a silicon-controlled rectifier (SCR) can be used to switch on a DC motor, and how the SCR is switched off again. (4)

1.3 Explain the function of an inductor and give ONE application of it. (3)

1.4 Draw a labelled block diagram of a microcontroller-based control system. (5)

1.5 Define the term amplitude. (2)

[20]

QUESTION 2

2.1 Name FIVE advantages of electric drives. (5)

2.2 Explain how a servo motor uses feedback to reach and hold a set position. (4)

2.3 Draw a labelled circuit diagram of an operational amplifier used as a proportional (P) controller (inverting amplifier), and explain how its gain is set. (6)

2.4 The proportional controller in QUESTION 2.3 has a feedback resistor of 100 kΩ and an input resistor of 10 kΩ. Calculate the gain and the output voltage for an input of 0,5 V. (3)

2.5 Define the term programming. (2)

[20]

QUESTION 3

3.1 Explain the purpose of a counterbalance valve. (3)

3.2 List FIVE things (parameters) that must be considered when selecting a hydraulic cylinder. (5)

3.3 Draw the symbol of a hydraulic accumulator and explain its function. (4)

3.4 Design a hydraulic circuit for a log splitter. Make use of: a double-acting cylinder; a manually operated 4/3-way valve with spring centring and a tandem centre; a pressure relief valve; a pressure gauge; and a filter in the return line. (8)

[20]

QUESTION 4

4.1 Explain the difference between a pressure relief valve and a pressure-reducing valve. (4)

4.2 Explain what an electrical interlock is, and why it is used in a circuit with a double-solenoid valve. (3)

4.3 A double-acting cylinder is controlled by a 4/2-way double-solenoid valve (Y1 extends, Y2 retracts). Design the electrical control circuit so that: push button S1 extends the cylinder; push button S2 retracts it; Y1 and Y2 can never be energised at the same time (interlock); an emergency stop removes the power; and a lamp H1 lights up when limit switch S3 detects that the cylinder is fully extended. (11)

4.4 Name TWO types of sensors that can detect the end positions of a cylinder. (2)

[20]

QUESTION 5

5.1 Explain the purpose of the isolator when faulty components must be replaced. (3)

5.2 Explain how air can get into a hydraulic system and what its effects are. (4)

5.3 A hydraulic pump makes a loud noise while it runs. Give THREE possible causes. (3)

5.4 Explain, step by step, the procedure for replacing a hydraulic pump. (7)

5.5 Explain the purpose of pilot lights (indicator lamps) on a control panel, and give ONE example. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER